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Systems and Methods For Refurbishment Of Motor Generators

Abstract

Systems and methods are provided to refurbish a commutator in-place without disassembly of a motor generator. The disclosed system may include a drive/brake system for controlling the rotation of the commutator during the refurbishment process, and a multi-axis Computer Numerical Control (CNC) motion and machining platform for controlling processing and refurbishment devices that traverse the commutator surface. Such devices may include an undercutting device to remove dielectric material from between commutator bars, a stoning device to obtain a polished surface finish for the commutator, and a chamfering device to remove burrs at the edges of the commutator grooves left by the stoning process. Alignment and inspection devices may be utilized along with mobile panels and cameras. The subsystems and components of the disclosed system may comprise a modular implementation designed for transportation and assembly in space-constrained environments.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This non-provisional patent application claims priority to, and incorporates herein by reference in its entirety, U.S. Provisional Patent Application No. 63/428,395 that was filed Nov. 28, 2022.

FIELD OF THE DISCLOSURE

[0003] The present disclosure relates in general to the field of refurbishment methods for components of motor generators (MG), such as commutators and other electrical rotors such as slip rings, and in particular to systems and methods and apparatuses for undercutting, stoning/polishing, chamfering/deburring and inspecting MG commutators using a robotic control system and user interface.

BACKGROUND

[0004] Basic techniques for the manual refurbishment of motors are known in the art. Past methods for the manual refurbishing motors, however, are time-consuming and produce inconsistent results. Improved solutions are desired for automating the refurbishment processes of large motors. Features of the present disclosure overcome various deficiencies of the prior art by providing a method, system and apparatus having advantages that will become apparent from the following disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The foregoing and other objects, features, and advantages for embodiments of the present disclosure will be apparent from the following more particular description of the embodiments as illustrated in the accompanying drawings, in which reference characters refer to the same components throughout the various views. The drawings are not necessarily to scale; emphasis instead being placed upon illustrating principles of the present disclosure.

[0006] FIG. 1 illustrates an exemplary implementation of a refurbishment system connected to a motor generator, in accordance with certain embodiments of the present disclosure.

[0007] FIG. 2 illustrates the subcomponents of the refurbishment system shown in FIG. 1, in accordance with certain embodiments of the present disclosure.

[0008] FIG. 3 is a flowchart illustrating exemplary steps of a method for operating a refurbishment system, in accordance with certain embodiments of the present disclosure.

[0009] FIG. 4 illustrates an exemplary drive-brake assembly of a refurbishment system, in accordance with certain embodiments of the present disclosure.

[0010] FIG. 5 illustrates an exemplary CNC machining/motion platform of a refurbishment system, in accordance with certain embodiments of the present disclosure.

[0011] FIGS. **6A-6C** illustrate front, side and prospective views of an exemplary laser alignment device for the CNC machining/motion platform shown in FIG. 5, in accordance with certain embodiments of the present disclosure.

[0012] FIGS. **7A-7C** illustrate front, side and prospective views of an exemplary inspection device for the CNC machining/motion platform shown in FIG. 5, in accordance with certain embodiments of the present disclosure.

[0013] FIGS. **8A-8D** illustrate front, side, prospective and cross-sectional views of an exemplary undercutting device for the CNC machining/motion platform shown in FIG. 5, in accordance with certain embodiments of the present disclosure.

[0014] FIGS. **9A-9C** illustrate front, side and prospective views of an exemplary stoning device for the CNC machining/motion platform shown in FIG. 5, in accordance with certain embodiments of the present disclosure.

[0015] FIGS. **10A-10C** illustrate front, side and prospective views of an exemplary chamfering device for the CNC machining/motion platform shown in FIG. 5, in accordance with certain embodiments of the present disclosure.

[0016] FIG. **11** illustrates subcomponents of an exemplary **110** VAC control box housed with a carrying case, in accordance with certain embodiments of the present disclosure.

BRIEF SUMMARY OF THE DISCLOSURE

[0017] The following presents a simplified summary of the disclosure in order to provide a basic understanding of some aspects of the disclosure. This summary is not an extensive overview of the disclosure. It is intended neither to identify key or critical elements of the disclosure, nor to delineate the scope of the disclosure. Its sole purpose is to present some concepts, in accordance with the disclosure, in a simplified form as a prelude to the more detailed description presented herein.

[0018] According to one technical advantage, the present disclosure may provide significant improvements over existing equipment by enabling the refurbishment of a commutator in-place without significant disassembly of the motor generator. For example, when the commutator of a motor generator needs to be refurbished, it may be necessary to perform complex rigging operations to disassemble the motor generator and then extract the entire rotor and send the rotor to a machine shop to be manually refurbished. These operations require that the vehicle be stationed and secured in-place prior to disassembly, and additional time and manpower would be required for rigging and preparation of the disassembly process. In certain embodiments, the presently disclosed motor-generator refurbishment system may only require removal of one inspection window cover-plate from the motor generator in order to perform a complete refurbishment in-situ. Accordingly, a benefit of the present disclosure may include the reduction of significant cost, overhead and time for the refurbishment of motors. In addition, by opening only one access port using the disclosed methods, the motor generator may be exposed to less contamination in the environment. It may therefore be possible to eliminate or reduce some of the environmental provisions currently employed, such as erecting tents or halting work in the vicinity of the machine days in advance.

[0019] Another advantage of the presently disclosed MG refurbishment system may include its programmable and repeatable performance. The disclosed system may ensure a repeatable process based on set parameters based on the disclosed automated and/or robotic processes. Additionally, in some embodiments, the system may be programmed to compensate for differences in height along the axis of a cylindrical component and return it to a nominally concentric shape, i.e. removing taper from commutator. The controlled motion of refurbishment or processing devices, as provided by a motion platform of the disclosed system, may generate a superior uniformity of undercutting, resurfacing and chamfering. In certain embodiments, an operator may still be provided with control over the process via an user interface. Similarly, the drive/brake unit may be controlled by utilizing varying rotational speeds for stoning. Pre-existing methods and techniques fail to provide such capabilities. For all operations, in accordance with certain embodiments, the software

implementation for the disclosed systems and methods may allow an operator to define linear speeds, rotational speeds, cut depths, and other specific parameters while also providing the option to run sequences of pre-planned coordinated motions.

[0020] In accordance with certain embodiments, the disclosed system may also comprise non-binding undercut and chamfering devices. Conventional tools are handheld and implemented manually, and are not applicable to a rigidly mounted implementation. The presently disclosed chamfering and undercut devices are designed with spring-loaded and self-correcting geometries that ensure the disclosed devices maintain cutting pressure and do not become misaligned or overloaded causing breakage. Additionally, in some embodiments, a laser alignment device may be utilized to project a visual aid on the commutator to assist with the installation of refurbishment devices and/or the inspection of the refurbishment process. In certain embodiments, an inspection device may be used to gauge commutator heights and/or the distance and alignment of a commutator bar relative to a refurbishment device mounted on the motion platform.

[0021] The presently disclosed systems, devices and methods may provide any one of the following benefits: a reduction in the time and costs for disassembly; an increased speed of refurbishment operations; a more consistent and repeatable process; a reduction in the number of operators required; a decrease in environmental controls; increased portability; a reduction in operator fatigue, risks and accidents; less rework from human error; and, less skill/talent required for acceptable results. In some embodiments, the present disclosure may result in a higher production yield over prior methods and techniques. Further advantages and features of the present disclosure are illustrated in the drawings and described in detail below.

DETAILED DESCRIPTION

[0022] The present disclosure comprises a comprehensive in-situ refurbishment system **100** for the refurbishing of components of motor-generator sets or motor generators (MG) **1** that may include electrical rotor components such as slip rings and commutators **2**, as shown in FIG. **1**. In accordance with certain embodiments, the system **100** may include subsystems comprising: a multi-axis Computer Numerical Control (CNC) motion and machining platform **110** internally mounted within the motor generator **1**; a drive/brake system **130** mounted on a DC end **5** of the motor generator **1** for spinning, stopping, and indexing a rotor **10**; and, industrialized robotic control systems **160**, **170** and user interface **150** that may include hardware and/or software for controlling the refurbishment system **100**. The subsystems of the system **100** may comprise a modular implementation designed for transportation, assembly and use in space-constrained environments, such as ships and other large vehicles. Such subsystems may be adapted to be transported within carrying cases **180** that house the control systems **160**, **170**.

[0023] FIG. **1** is a cutaway diagram, depicting a cutout **9** in the housing **8** of a motor generator **1** for illustration purposes, that shows the internal position of the machining platform **110** within the motor generator **1** in accordance with certain embodiments. In order to be internally installed, the machining platform **110** may be inserted through a window **7** in the housing **8** of a motor generator **1**, and may be mounted on a brush holder **3** after the removal of the brushes **4** (not shown). The brushes **4** may comprise sliding contacts that make electrical connections with the rotating part or rotor **10** of a motor generator **1**, which may include the commutator bars **2** that rotate on a shaft within a motor generator **1**. For further clarity without the visual obstruction of the components of the motor generator **1** shown in FIG. **1**, FIG. **2** illustrates the structural relationship of the presently disclosed subsystems of the refurbishment system **100** as positioned and installed on, within and adjacent to the motor generator **1** as shown in FIG. **1**.

[0024] In some embodiments, the system **100** may perform multiple operations applicable to refurbishing large motors **1** that may comprise the following steps for a presently disclosed method **300**: removing dielectric material between commutator bars, i.e. undercutting [block **301**]; bulk removal and refinishing of a commutator surface, i.e. stoning and polishing [block **302**]; chamfering and deburring commutator bars, i.e. chamfering [block **303**]; and inspecting

commutator bar heights, the distance between a commutator bar **2** and a processing/refurbishment tools/device **140** mounted on the motion platform **110**, and/or the alignment of the refurbishment device **140** relative to the distance of a commutator bar **2**, i.e. inspection [block **304**], as shown in the flowchart illustrated in FIG. **3**. The present disclosure may be implemented to resurface and recondition MG commutators **2** without major disassembly of the motor generator **1** and performing the operations (e.g. undercutting, stoning/polishing, and chamfering) of the disclosed method **300** in-place in a space-constrained environment. Components of the presently disclosed refurbishment system **100** may be installed through one or more preexisting inspection windows **7** in the housing **8** of a motor generator **1**, and operated via external controls comprising the robotic control systems **160**, **170** and user interface **150**.

[0025] As may be appreciated by those skilled in the art, motor refurbishment is required in many industries. Commutator refurbishment has been a regular maintenance task for vessels equipped with motor generators **1**, which was manually performed. Prior methods for refurbishing a commutator **2** included time and labor intensive tasks, especially when performed in limited space environments. It is necessary to maintain the functionality of a motor generator **1**, which is extremely sensitive to environmental factors and becomes contaminated naturally over its lifetime. The most frequent refurbishment operation that must be performed is stoning, which comprises the refinishing/resurfacing the surface **2'** of a commutator **2** by removing a contaminated exterior from the commutator surface **2'**. Historically, this operation was manually performed without removing the MG set **1** from the ship, but with significant disassembly which required the implementation of extensive environmental controls to maintain cleanliness and safety. In certain circumstances, there may be a limit to the number of times the commutator **2** can be stoned without performing additional refurbishment operations such as undercutting and chamfering which typically require a significantly greater amount of time and labor when manually performed.

[0026] The undercut operation usually involves cutting the dielectric material present between commutator bars **2** to a specified level below the commutator bar surfaces **2'**. A commutator **2** may comprise copper bars or other mechanical segments/devices arranged to reverse armature connections in motor generator sets **1** at the proper instant so that current continues to flow in only one direction, as the commutator bars **2** may be separated from each other by mica insulation that may be cut at a height below the copper segments/bars **2** permitting the commutator **2** to change the generated AC voltage to a pulsating DC voltage. The chamfering operation involves deburring the edges of each commutator bar **2** following undercutting and stoning. Typically, the stoning operation of prior technics involves installing a manually-controlled slide with abrasive grinding stones into the motor generator **1**. A skilled technician then installs a rotary drive-interface such as an air motor to one end of the motor generator **1** to achieve a desired commutator rotation speed. When the commutator **2** is up to speed, the technician manually drives stones into the surface and across the commutator **2** to remove dielectric material. The operation requires a progression from coarse to fine stones, with another spin-down and spin-up operation required between each new stone grit. The speed of the commutator rotation, speed of stoning, and amount of stone to infeed during this operation are usually evaluated by an operator who can recognize the expected and/or desired sound during refurbishment. The speed of the commutator rotation is not effectively controllable by the operator as the speed is usually directly dependent on the pressure of the available air connection. The lack of an effective braking mechanism for the commutator **2** significantly increases the time of spin-down between each stone grit change.

[0027] Such stoning, undercutting, and chamfering conventionally require rigging operations to disassemble the motor generators **1**, which adds a significant overhead to the operation and is especially difficult to achieve in a space-constrained environment. Prior technics lacked the disclosed system **100** to perform undercutting and chamfering operations without major disassembly of motor generators **1**. In comparison with historical data, the refurbishment of certain MG sets **1** utilizing the present disclosure reduced the refurbishment time by at least 18%. Prior

technics uncontrolled drive-and-brake functionality by way of pneumatic or electric motors with manual mechanical or hydraulic braking, which lack the abilities provided by the present disclosure such as: rotor speed control, indexing, position feedback, torque feedback, rapid braking, and safety interlocks. Previous multi-axis stoning manual platforms also lack the advantages provided by the present disclosure, including: computer-controlled and simultaneous motion in multiple axes, safety interlocks, position feedback and inspection capabilities, provisions for mounting tooling other than stones, camera feedback, or complete refurbishment operations controlled through an inspection window **7** in the housing **8** of a motor generator **1**.

[0028] In accordance with certain embodiments, the present disclosure may be intended to refurbish motor-generators **1** in-place in a space-constrained environment. The disclosed system **100** may comprise the subcomponents shown in FIGS. **1-2** and **4-11**, including: a drive-brake assembly **130**, comprising a drive servo-motor **137** and pneumatic brake **138**; power conditioning and logic hardware **160, 170**; one or more mobile panels **150**, comprising user interfaces **150** and touchscreen displays or monitors **150'**; one or more motion platforms **110**; processing/refurbishment tools/devices **140**; one or more sets of cameras **125, 126** monitoring the refurbishment process; and, a modular form factor for such subcomponents adapted for commercial and military transportation, as described below, such that the disassembled subcomponents may be packaged as kits within one or more carrying cases **180**.

[0029] As a typical example, in accordance with certain embodiments, use of the presently disclosed MG refurbishment system **100** to perform a complete motor-generator refurbishment may comprise of the following process. First, the motor generator **1** must be prepared. The multiple storage cases **180** of the containerized system **100** may be carried by hand into the limited space environment of an engine room where the motor-generator sets **1** operate to power a vessel. Here, technicians may open one access panel or inspection window **7** of the MG **1** and remove several brushes **4** and brush holders **3**. Any dust cover may also be removed from the front of the MG **1**, in order to allow for installation of the drive-brake assembly **130**. The diameter of the commutator **2** may be measured with a pi tape.

[0030] Next, the drive-brake assembly **130** may be installed. In accordance with certain embodiments, as shown in FIG. **4**, the drive-brake assembly **130** may comprise an end bell **131**, an end bell adapter **131'**, a motor plate **131''**, a brake encoder **132**, a shaft assembly **133**, a shaft **134**, a spider coupling **135**, a multi-jackbolt tensioner (MJT) or Superbolt™ clamping/bolting assembly **136**, a drive servo-motor **137**, a pneumatic brake **138**, and an electro-pneumatic pressure regulator **139**. The drive/brake assembly **130** may spin the rotor **10** together with the commutator **2** during stoning operations, and stop the rotation when stoning is complete. The drive/brake assembly **130** may also be used to iterate between commutator slots **2** for the undercut/chamfer operations. The drive portion of the assembly **130** may spin the commutator **2** by use of the servo motor **137**. The brake portion of the assembly **130** may utilize a pneumatic brake **138** adapted to connect to a shop air connection. In order to attach the drive/brake assembly **130** to a motor generator **1**, an end bell **131** may be utilized without an end bell adapter **131'** as shown in FIG. **1**.

[0031] In some embodiments, referring back to FIG. **4**, a backside of an end bell **131** may be attached to an end bell adapter **131'** in order to properly space the drive-brake assembly **130** relative to the DC end **5** of the motor generator **1**. In end embodiment, the end bell **131** and the end bell adapter **131'** may have holes spaced to correspond to bolt holes existing in the DC end **5** such that bolts may fasten the end bell **131** and the end bell adapter **131'** to the motor generator **1**. Prior to attaching the motor plate **131''** to the front side of the end bell **131**, the shaft assembly **133** may be installed at a predetermined height relative to the front side of the end bell **131**.

[0032] For example, the shaft **134** of the shaft assembly **133** may be threaded to correspond to threads within the end of the rotor **10**. The shaft **134** may be threaded into rotor end **10'** at a desired depth, and the Superbolt™ assembly **136** may be attached to the shaft collar **134'**, such that the Superbolt™ assembly **136** is even with the front side of the end bell **131**, in accordance with

certain embodiments. The spider coupling **135** may be mounted on the Superbolt™ assembly **136** in order to provide an effective power transmission without torsional oscillation/vibration and to protect the subcomponents of the drive/brake assembly **130**. In some embodiments, a gauge may be used to confirm when the shaft assembly **133** is installed within a desired height range, based on the height of the installed end bell **131** and/or the end bell adapter **131'**, such that the height of the spider coupling **135** may be adjusted for proper engagement with connectors of the drive servo-motor **137** and pneumatic brake **138** that are mounted on the exterior side of the motor plate **131''**. The motor plate **131''** may be mounted to the front side of the end bell **131** once the shaft assembly **133** is installed at a suitable height to ensure such a proper engagement upon mounting the motor plate **131''**. In end embodiment, a desired air pressure may be maintained via a brake encoder **132**, which may take pressure readings from the brake system and generate corresponding pressure values, together with an electro-pneumatic pressure regulator **139** which may maintain air pressure in proportion to an electric signal.

[0033] In certain embodiments, the motion platform **110** may then be installed onto an existing interface on the brush holder/arm **3** as shown in FIG. **1**. In some embodiments, the motion platform **110** may be adapted to align with the commutator slots **2** mounted on the rotor **10**. Referring to FIG. **2**, multiple cameras **125**, **126** may be installed in order to provide feedback to the operator during operations, especially operations where refurbishment devices **140** must align to the commutator slots **2**. In some embodiments, an overhead camera **125** may provide the user with an overview of refurbishment operations performed by the disclosed system **100** within the motor generator **1**. In certain embodiments, a slot camera **126** may provide the user with a view of the commutator slots **2** and the tips of the processing device **140** during the refurbishment process. One or more vacuum ducts **122** may be mounted on other brush arms **3** in order to recover debris emitted during refurbishment. The vacuum ducts **122** may be adapted to connect to vacuum hoses **123** (not shown), which may be connected to vacuums.

[0034] In accordance with certain embodiments, as shown in FIG. **5**, the CNC machining/motion platform **110** may comprise a slide mount **111** adapted to attach to the brush holder/arm **3** of the motor generator **1** shown in FIG. **1**. Referring back to FIG. **5**, the slide mount **111** may have bolt holes adapted to receive mounting studs/bolts/fasteners **111'** configured to correspond with preexisting bolt holes in the brush holder **3**, and further configured to receive bolts nuts **111''** for securing the slide mount **111** together with the motion platform **110** to the brush holder **3**. In some embodiments, studs/bolts **111'** and bolts nuts **111''** may secure the vacuum ducts **122** as shown in FIG. **2** to the brush arms **3** illustrated in FIG. **1**.

[0035] As shown in FIG. **5**, the motion platform **110** may further comprise a linear-traverse slide **112**, a traverse motor **113**, a gearbox **114**, a ball screw **115**, a traverse carriage **116**, hard stops **117**, an infeed slide **118**, a worm gear **119**, an infeed motor **120**, and an energy chain **121**. In certain embodiment, the motion platform **110** may move a predetermined set of processing/refurbishment devices **140** that may be adapted for the undercutting, stoning and polishing, chamfering and deburring, and inspecting of commutator bars **2** as describing herein. Such refurbishment devices **140** may be mounted on an infeed slide **118** moved by a worm gear **119** controlled by an infeed motor **120**. The infeed slide **118** may move the refurbishment device **140** toward the commutator **2** or slot surfaces **2'**. In some embodiments, an energy chain **121** may comprise an infeed cable that may provide power to the infeed motor **120** and may transmit limit-switch signals that limit movement of the infeed slide **118**. The infeed slide **118** may be mounted to the traverse slide **112** via a traverse carriage **116**, such that the traverse slide **112** guides the traverse carriage **116** along a linear axis **13** that traverses the longitudinal axis of the motion platform **110**. By loosening the mounting bolts **111'**, the traverse slide **112** may be moved to align the linear axis **13** with the commutator slots **2**. In certain embodiments, the linear axis **13** corresponds to the longitudinal axis of the commutator **2** and the motor generator **1** where the motion platform **110** is mounted parallel to the brush holders **3** that are parallel to the commutator **2** and the motor generator **1**. Accordingly,

the traverse carriage **116** may move along the linear axis **13** of the commutator **2**. In some embodiments, a traversing cable may provide power to the traversing motor **113** that controls the gearbox **114**, which may turn the ball screw **115** that drives the traverse carriage **116** along the traversing slide **112**. Hard stops **117** may be set to limit movement of the traversing slide **112**. [0036] Installation of the refurbishment system **100** may include the step of connecting the subsystems via system cables **101**, including the user interface **150** and control boxes **160**, **170** so that the technician may initiate the generation of system commands to begin the refurbishment process. The system **100** may be used to perform the abovementioned operations, such as undercutting **301**, stoning **302**, and chamfering **303**. In certain embodiments, the operator/technician initiates the generation of commands through the user interface **150** to align the refurbishment devices **140**, control the depth of cut, and perform a predetermined number of cuts or passes. Such predetermined subroutines may generate the issuance of specific motion commands to the motion platform **110** and the drive/brake system **130** in order to accomplish the refurbishment process. The subroutines may impose limits that may automatically preclude the refurbishment device/tool **140** from crashing or damaging the commutator **2**. The operator may review and confirm each set of actions in order further prevent damage and ensure that the refurbishment process is performed.

[0037] In accordance with certain embodiments, various processing/refurbishment devices **140** may be adapted to be mounted and dismounted on the infeed slide **118** to perform the varying processes of the presently disclosed method **300** described herein. In some embodiments, the devices **140** may be equipped with two or more guide pins and/or one or more screws/bolts **140'** as described herein that correspond and fit holes in the infeed slide **118** that may be adapted to receive such pins/screws/bolts. This mounting and locking mechanism may be reversible such that various devices **140** may be selected from a group of refurbishment/processing devices **140** to be used at varying times during the refurbishment process in a varying order. For example, a visual and/or alignment inspection may be conducted before and/or after any of the undercutting **301**, stoning **302**, and chamfering **303** processes. In an embodiment, the structure of such a mounting and locking mechanism may be reversed such that the devices **140** may have holes adapted to receive pins/screws/bolts that extend from the infeed slide **118**. One or more magnets may be embedded in the devices **140** and/or the infeed slide **118** in order to generate a magnetic attraction between the two components strong enough to keep the devices **140** in place while mounting and securing the devices **140** unto the infeed slide **118**.

[0038] In certain embodiments, the group of refurbishment/processing devices **140** that may be mounted on the motion platform **110** may include a laser alignment device **141** as depicted in FIGS. 6A-6C. The laser alignment device **141** may be utilized to project a visual aid on the commutator **2**. In some embodiments, the laser alignment device **141** may project a laser crosshair pattern to be used to align the motion platform **110** to the commutator slots **2**. Alignment adjustments may be made by loosening fasteners **111'** that secure subcomponents of the motion platform **2**, repositioning the subcomponents relative to each other, and retightening the fasteners **111'**. As shown in FIG. 6A, the laser alignment device **141** may comprise a switch **141''** to activate/deactivate the alignment laser crosshair **149** shown in FIG. 6C. In an embodiment, the laser alignment device **141** may comprise a laser projector **141'**.

[0039] In some embodiments, the laser alignment device **141** may comprise a slot alignment device **141** that may be mounted upon installation of the motion platform **110** within the motor generator **1** as shown in FIG. 1. The slot alignment device **141** may verify the proper alignment of the motion platform **110** based on the location of the commutator slots **2**. A method for installing the slot alignment device **141** may include the step of placing the slot alignment device **141** firmly on the infeed slide **118** in an orientation corresponding to the pins/screws/bolts. The method may include verifying the device **141** is flush so that the magnetic attraction is strong enough to keep the device **141** in place. In some embodiments, the method may include the steps of turning on the slot

alignment device **141** to activate laser crosshairs by flipping a switch **141''**, and jogging the infeed slide **118** towards and/or away from the commutator surface **2'** to focus the laser crosshairs. The laser crosshairs may be further focused by turning a knurled head **141'''** of the laser diode as shown in FIG. 6A. In an embodiment, the alignment device **141** may be held in place via the aforementioned magnetic attraction without use of any pins/screws/bolts. The alignment device **141** may be powered by an internal battery, such as a CR2032 battery, and the battery may be replaced by removing a cover **141''''** from the housing of the alignment device **141**.

[0040] In certain embodiments, the group of refurbishment/processing devices **140** may include an inspection device **142** as depicted in FIGS. 7A-7C. The inspection device **142** may be utilized to gauge commutator heights and/or the alignment of the motion platform **110** relative to the commutator **2**. In an embodiment, the inspection device **142** may ensure that the motion platform **110** is positioned at a predetermined distance from the commutator **2**. The motion platform **110** may comprise an inspection tip or pin **142'** that may be pushed inward upon contacting a commutator bar **2** and receiving a biasing force, resulting in a pressure gauge **142''** measuring the biasing force that may be recorded and displayed on a pressure gauge face or display **142'''**. A method for installing the inspection device **142** may include the step of mounting the device **142** on the infeed slide **118** by aligning and tightening the pins/screws/bolts **140'**.

[0041] The group of refurbishment devices **140** may include an undercutting device **143** as depicted in FIGS. 8A-8D. In some embodiments, the undercutting device **143** may be adapted to cut and remove mica between the commutator bars **2**. The undercutting device **143** may comprise one or more blades **143'**, which may be replaceable in accordance with certain embodiments. As shown in FIG. 8D, the blades **143'** may be adapted to receive a biasing force from a biasing mechanism that may comprise preloaded biasing pins **143''**, springs **143'''** and bolts **143''''**. A method for installing the undercutting device **143** may include the step of mounting the device **143** on the infeed slide **118** by aligning and tightening the pins/screws/bolts **140'**. To replace the blades **143'**, the preload biasing pins **143''**, springs **143'''**, and bolts **143''''** may be removed. Screws and or hinge pins **143'''''** holding the blades **143'** may then be removed. The blade **143'** may subsequently be removed, and a replacement blade **143'** may be installed by the repeating the aforementioned process in reverse.

[0042] In accordance with certain embodiments, the undercutting **301** process may utilize a drive/brake indexing function of the drive/brake assembly **130**, the cameras **125/126**, the undercutting device **143**, and the motion platform **110** to remove dielectric material from between the commutator bars **2**. A method **800** for undercutting **301**, e.g. using the undercutting device **143** shown in in FIGS. 8A-8D, may comprise the following steps: setting an initial tool height relative to the commutator surface **2'** using a paper gauge; setting a desired cutting force or bias based on a predetermined set of parameters for a preload and per-cycle infeed phase, wherein such preload and per-cycle infeed parameters may be defined by predetermined instructions programmed in software and/or configured to be executed by a processor adapted to control operations of the presently disclosed refurbishment system **100**; setting—via predetermined instructions—a predetermined number of undercutting passes, e.g. the number of times that the motion platform **110** moves the undercutting device **143** back and forth along the linear axis **13** of the commutator **2**; setting—via predetermined instructions—an undercutting time-period for undercutting **301**; executing the aforementioned instructions, e.g. by running the software, via the processor adapted to control operations of the refurbishment system **100**; evaluating the new commutator slot surface **2'** depth, e.g. by gauging the commutator heights via the inspection device **142** as described herein, and determining whether additional undercutting passes are required when the commutator surface **2'** do not inwardly push the biasing pins **143''** with a predetermined force; when the undercut process removes obstructing dielectric material from the commutator bars **2** such that the distance between the motion platform **110** and a commutator bar **2** is reduced, permitting the commutator surface **2'** to inwardly push the biasing pins **143''** with a predetermined force, then an indication that the

undercut process **301** has completed may be recorded and the drive/brake assembly **130** may rotate the rotor **10** along the rotor axis **12** by a predetermined iterative amount so that the motion platform **110** may engage the next, adjacent row/area of commutator slot surface **2'** in order to repeat the aforementioned undercutting process **301**; validating that the undercutting device **143** is aligned to the next commutator slot surface **2'** prior to commencing the undercutting process **301**; and, repeating the aforementioned undercutting process **301** for the entire commutator slot surface **2'**. [0043] The group of refurbishment devices **140** may include a stoning device **144** as depicted in FIGS. **9A-9C**. In some embodiments, the stoning device **144** may be utilized to obtain a predetermined geometry and surface finish for the commutator **2**. A method for installing the stoning device **144** may include the step of mounting the device **144** on the infeed slide **118** by aligning and tightening the pins/screws/bolts **140'**. The stoning device **144** may be adapted to receive various stoning bars **144'** having varying textures, such as extra course (EC), course (C), medium (M), fine (F), polish (P) and extra polish (EP). Such stoning bars **144'** may be replaceable. As shown in FIG. **9A**, the stoning bars **144'** may be attached to the stoning device **144** via mounting braces **144''** secured by mounting bolts or screw **144'''**. To replace a stoning bar **144'**, the mounting braces **144''** may be loosened and tightened again after another stoning bar **144'** is set in place. The height of the stoning bar **144'** may be set to a predetermined limit.

[0044] In certain embodiments, the stoning **302** process may utilize the drive/brake assembly **130** to spin the commutator **2** and then use the motion platform **110** to move the stoning bar **144'** toward the commutator **2** at predetermined speeds. Stoning bars **144'** having varying textures may be applied to the commutator **2** in a predetermined order, and processing and controls parameters may be set in a corresponding order. Such parameters may include predetermined speeds and infeed patterns or profiles. For example, the infeed profiles may comprise a square pattern and saw pattern. The infeed profiles may be defined by predetermined instructions programmed in software and/or configured to be executed by a processor adapted to control operations of the presently disclosed refurbishment system **100**. A method **900** for stoning **302**, e.g. using the stoning device **144** shown in in FIGS. **9A-9C**, may comprise the following steps: spinning the commutator **2** up to a predetermined speed; setting an initial height for the stoning device **144** relative to the commutator surface **2'** using cameras **125/126** to visually set the stone height near the surface **2'**; recording the starting height via predetermined instructions; setting a predetermined maximum processing depth via predetermined instructions, e.g. a maximum difference between the starting height and a measured height received during the stoning process **302**; setting—via predetermined instructions—a predetermined infeed pattern and per-pass infeed amount; executing the aforementioned instructions, e.g. by running the software, via the processor adapted to control operations of the refurbishment system **100**, and monitoring the performance of the stoning process **302** based on the received measured heights; upon reaching the maximum processing depth, stopping the rotation of the commutator **2** by utilizing the drive/brake assembly **130**; evaluating the condition of the commutator **2**, e.g. by using the cameras **125/126** to visually determine the grit of the commutator surface **2'**; replacing the stoning bar **144'** with another stoning bar **144'** having a different texture; and, repeating the aforementioned stoning process **302** with the other stoning bar **144'** for the entire commutator surface **2'**.

[0045] The group of refurbishment devices **140** may include a chamfering device **145** as depicted in FIGS. **10A-10C**. In some embodiments, the chamfering device **145** may be utilized to remove the burrs at the edges of the commutator grooves left by the stoning process **302**. A method for installing the chamfering device **145** may include the step of mounting the device **145** on the infeed slide **118** by aligning and tightening the pin/screw/bolt **140'**. The chamfering device **145** may be adapted to hold spring-loaded chamfering inserts **145'** mounted via insert screws **145''** onto the insert springs **145'''**. The chamfering inserts **145'** may be replaced by loosening the insert screws **145''**.

[0046] In certain embodiments, the chamfering process **303** may utilize a drive/brake indexing

function of the drive/brake assembly **130** along with cameras **125/126**, the motion platform **110**, and the chamfering device **145** to align to a commutator slot surface **2'** and deburr the edges of the commutator bars **2** along each slot. A method **1000** for chamfering **303**, e.g. using the chamfering device **145** shown in in FIGS. **10A-10C**, may comprise the following steps: setting an initial height for the chamfering device **145** relative to the commutator slot surface **2'** using a paper gauge; setting a predetermined cutting force by setting a preload bias via the insert springs **145'''** for the chamfering device **145**; setting—via predetermined instructions—a predetermined number of chamfering passes, e.g. the number of times that the motion platform **110** moves the chamfering device **145** back and forth along the linear axis **13** of the commutator **2**; executing the aforementioned instructions, e.g. by running the software, via the processor adapted to control operations of the refurbishment system **100**; monitoring and evaluating the performance of the chamfering process **303**; e.g. by using the cameras **125/126** to visually determine the grit of the edges of the commutator surface **2'** and determining whether to conduct additional chamfering passes; when the chamfering process removed the visible burrs at the edges of the commutator grooves, then an indication that the chamfering process **303** has completed may be recorded and the drive/brake assembly **130** may rotate the rotor **10** along the rotor axis **12** by a predetermined iterative amount so that the motion platform **110** may engage the next, adjacent row/area of commutator slot surface **2'** in order to repeat the aforementioned chamfering process **303**; validating that the chamfering device **145** is aligned to the next commutator slot surface **2'** prior to commencing the chamfering process **303**; and, repeating the aforementioned chamfering process **303** for the entire commutator slot surface **2'**.

[0047] In some embodiments, the present disclosure may be implemented modularly for transportation purposes in order to carry the subcomponents of the refurbishment system **100** to space-constrained environments. For example, in some embodiments, power conditioning and logic hardware may be stored and transported within one or more carrying cases **180**. As shown in FIG. **11**, such a carrying cases **180** for the **110** VAC control box **160** may house power-conditioning equipment, a battery backup, an emergency power disconnect mechanism, and on/off button. In an embodiment, the carrying cases **180** for the **110** VAC control box **160** may contain a processor **190**, a computer device, and memory. The **110** VAC control box **160** may provide conditioned power to other subsystem components, such as the **24** VDC control box **170** and the drive servo-motor **137** for the drive-brake assembly **130**. In certain embodiments, referring back to FIG. **2**, the carrying cases **180** for the **24** VDC control box **170** houses a system controller that may control subsystems for the disclosed system **100** and communicate with the user interface **150**. The **24** VDC control box **170** may process and store input received through the mobile panels **150** and cameras **125**, **126**. In an embodiment, the carrying cases **180** for the **24** VDC control box **170** may contain a computer device, processors and memory. The **24** VDC control box **170** may control the cameras **125**, **126**, the mobile panel **150**, the motion platform **110**, and the drive/brake hardware of the drive/brake assembly **130**. The cameras **125**, **126** may include lights, which may also be controlled via the mobile panel **150**. The subsystems and/or subcomponents of the disclosed system **100** may connect and communicate with each other via the cables **101**. In an embodiment, subsystems and/or subcomponents may wirelessly communicate as described herein. For example, the mobile panel **150** may be adapted to wirelessly communicate with, and control, the motion platform **110**, the drive-brake assembly **130** and/or one or more of the cameras **125**, **126** and/or control boxes **160**, **170** as well as peripheral devices such as the vacuums connected to the vacuum ducts **122** via the vacuum hoses **123**.

[0048] The present disclosure may be applied to any rotating piece or member **2** of machinery **1**, such as a slip ring or commutator **2** of a motor-generator set **1**, motor **1**, or generator **1**. The drive-brake assembly **130** may be connected to such motor-generator sets **1**, motors **1**, or generators **1**. In certain embodiments, portions of the present disclosure may be applied to other rotating machinery **1** that require inspection or resurfacing, such as drive shafts or pumps.

[0049] This present disclosure may implement machine vision and/or machine learning to recognize features such as slots, scratches, or other defects. Inspection tools may be utilized to allow the disclosed system to measure diameters, map the contours of a rotating or stationary surface, probe heights, and/or make related measurements. These data could be used by the disclosed system to improve processing or externally by inspection and acceptance personnel.

[0050] In accordance with some embodiments, the system circuitry may include any combination of hardware, software, firmware, APIs, and/or other circuitry. The system circuitry may be implemented, for example, with one or more systems on a chip (SoC), servers, application specific integrated circuits (ASIC), field programmable gate arrays (FPGA), microprocessors, discrete analog and digital circuits, and other circuitry. The system circuitry may implement any desired functionality of the disclosed system. As just one example, the system circuitry may include one or more instruction processor and memory. The processor may be one or more devices operable to execute logic. The logic may include computer executable instructions or computer code embodied in the memory or in other memory that when executed by the processor, cause the processor to perform the features implemented by the logic. The computer code may include instructions executable with the processor. Logic, such as programs or circuitry, may be combined or split among multiple programs, distributed across several memories and processors, and may be implemented in a library, such as a shared library (e.g., a dynamic link library or DLL).

[0051] The memory stores, for example, control instructions for executing the features of the disclosed system. Examples of the memory may include non-volatile and/or volatile memory, such as a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM), or flash memory. Alternatively, or in addition, the memory may include an optical, magnetic (hard-drive) or any other form of data storage device. In one implementation, the processor executes the control instructions to carry out any desired functionality for the disclosed system, including without limitation those attributed to data/reflection receiver (e.g., relating to the data receiver circuitry), image generation, and/or profiler results generation. The control parameters may provide and specify configuration and operating options for the control instructions, and other functionality of the computer device.

[0052] The computer device may further include various data sources, as described herein. Each of the databases that are included in the data sources may be accessed by the system to obtain data for consideration during any one or more of the processes described herein. For example, the data receiver circuitry may access the data sources to obtain the information for generating the images and the reflection returns. In an embodiment, a data receiver circuitry may be configured to receive reflected signals.

[0053] All of the discussion, regardless of the particular implementation described, is exemplary in nature, rather than limiting. For example, although selected aspects, features, or components of the implementations are depicted as being stored in memories, all or part of the system or systems may be stored on, distributed across, or read from other computer readable storage media, for example, secondary storage devices such as hard disks, flash memory drives, floppy disks, and CD-ROMs. Moreover, the various modules and screen display functionality is but one example of such functionality and any other configurations encompassing similar functionality are possible.

[0054] The respective logic, software or instructions for implementing the processes, methods and/or techniques discussed above may be provided on computer readable storage media. The functions, acts or tasks illustrated in the figures or described herein may be executed in response to one or more sets of logic or instructions stored in or on computer readable media. The functions, acts or tasks are independent of the particular type of instructions set, storage media, processor or processing strategy and may be performed by software, hardware, integrated circuits, firmware, micro code and the like, operating alone or in combination. Likewise, processing strategies may include multiprocessing, multitasking, parallel processing and the like. In one embodiment, the instructions are stored on a removable media device for reading by local or remote systems. In

other embodiments, the logic or instructions are stored in a remote location for transfer through a computer network or over telephone lines. In yet other embodiments, the logic or instructions are stored within a given computer, central processing unit (“CPU”), graphics processing unit (“GPU”), or system.

[0055] In some embodiments, the computer device may include communication interfaces, system circuitry, input/output (I/O) interface circuitry, and display circuitry. The communication interfaces may include wireless transmitters and receivers (herein, “transceivers”) and any antennas used by the transmit-and-receive circuitry of the transceivers. The transceivers and antennas may support Wi-Fi network communications, for instance, under any version of IEEE 802.11, e.g., 802.11n or 802.11ac, or other wireless protocols such as Bluetooth, Wi-Fi, WLAN, cellular (4G, LTE/A). The communication interfaces may also include serial interfaces, such as universal serial bus (USB), serial ATA, IEEE 1394, lighting port, I2C, slimBus, or other serial interfaces. The communication interfaces may also include wireline transceivers to support wired communication protocols. The wireline transceivers may provide physical layer interfaces for any of a wide range of communication protocols, such as any type of Ethernet, Gigabit Ethernet, optical networking protocols, data over cable service interface specification (DOCSIS), digital subscriber line (DSL), Synchronous Optical Network (SONET), or other protocol.

[0056] While the present disclosure has been particularly shown and described with reference to an embodiment thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure. Although some of the drawings illustrate a number of operations in a particular order, operations that are not order-dependent may be reordered and other operations may be combined or broken out. While some reordering or other groupings are specifically mentioned, others will be apparent to those of ordinary skill in the art and so do not present an exhaustive list of alternatives. The presently disclosed instructions and code are examples, which may vary as understood by those skilled in the art, that are listed in order to illustrate the nature of certain embodiments.

Claims

1. A refurbishment system, comprising: a drive and brake system adapted to control the rotation of a rotating member; a motion platform adapted to traverse a portion of the rotating member; and, a refurbishment device adapted to be mounted on the motion platform, wherein the refurbishment device is adapted to engage the portion of the rotating member.
2. The system of claim 1, wherein the refurbishment device is selected from a group consisting of an undercutting device, a stoning device, and a chamfering device.
3. The system of claim 1, further comprising: an alignment device adapted to be mounted on the motion platform, the alignment device adapted to generate an alignment laser crosshair on the portion of the rotating member.
4. The system of claim 1, further comprising: an inspection device adapted to be mounted on the motion platform, the inspection device having a tip adapted to engage the portion of the rotating member.
5. The system of claim 1, wherein the rotating member comprises a commutator.
6. The system of claim 1, wherein the motion platform is adapted to be inserted through an aperture in a housing of a machine, wherein the rotating member is located within the machine.
7. The system of claim 6, wherein the machine is selected from a group consisting of a motor, a generator and motor-generator set.
8. The system of claim 7, wherein the motion platform is further adapted to be mounted on a brush holder within the machine, and wherein the drive and brake system is adapted to connect to an end of the machine.
9. The system of claim 8, further comprising: a vacuum duct adapted to be mounted on a second

brush holder within the machine.

10. The system of claim 1, further comprising: one or more cameras adapted to monitor the portion of the rotating member; and, a mobile panel comprising a display screen adapted to render images received from the one or more cameras, the mobile panel further comprising an user interface adapted to receive commands to control the motion platform and the drive and brake system.

11. The system of claim 10, further comprising: a processor adapted to execute instructions to control the motion platform and the drive and brake system based on commands received via the user interface.

12. A method for using a system of claim 2, comprising the steps of: removing, via the undercutting device, material from the portion of the rotating member; polishing, via the stoning device, the portion of the rotating member; and, deburring, via the chamfering device, the portion of the rotating member.

13. A method of claim 12, further comprising the steps of: generating, via an alignment device, an alignment laser crosshair, wherein the alignment laser crosshair is projected on the portion of the rotating member; and, aligning the motion platform based on the alignment laser crosshair.

14. The method of claim 12, wherein the removing step comprises undercutting dielectric material located on commutator slot surfaces between commutator bars, wherein the rotating member is a commutator comprising the commutator bars, the portion of the rotating member comprising the commutator slot surfaces between the commutator bars.

15. The method of claim 14, wherein the polishing step comprises stoning commutator bar surfaces, the portion of the rotating member further comprising the commutator bar surfaces.

16. The method of claim 15, wherein the deburring step comprises chamfering edges of the commutator bars, the portion of the rotating member further comprising the edges of the commutator bars.

17. A method of claim 16, further comprising the step of: inspecting, via an inspection device, a height of the commutator bars based on a biasing force measured by a pressure gauge of the inspection device.
